

Solution 3.3

$$\begin{aligned}\vec{u}_{\parallel} &= \frac{3 \cdot 1 + 2 \cdot (-1) + 2 \cdot 0}{1^2 + (-1)^2 + 0^2} \langle 1, -1, 0 \rangle \\ &= \frac{1}{2} \langle 1, -1, 0 \rangle \\ &= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle\end{aligned}$$